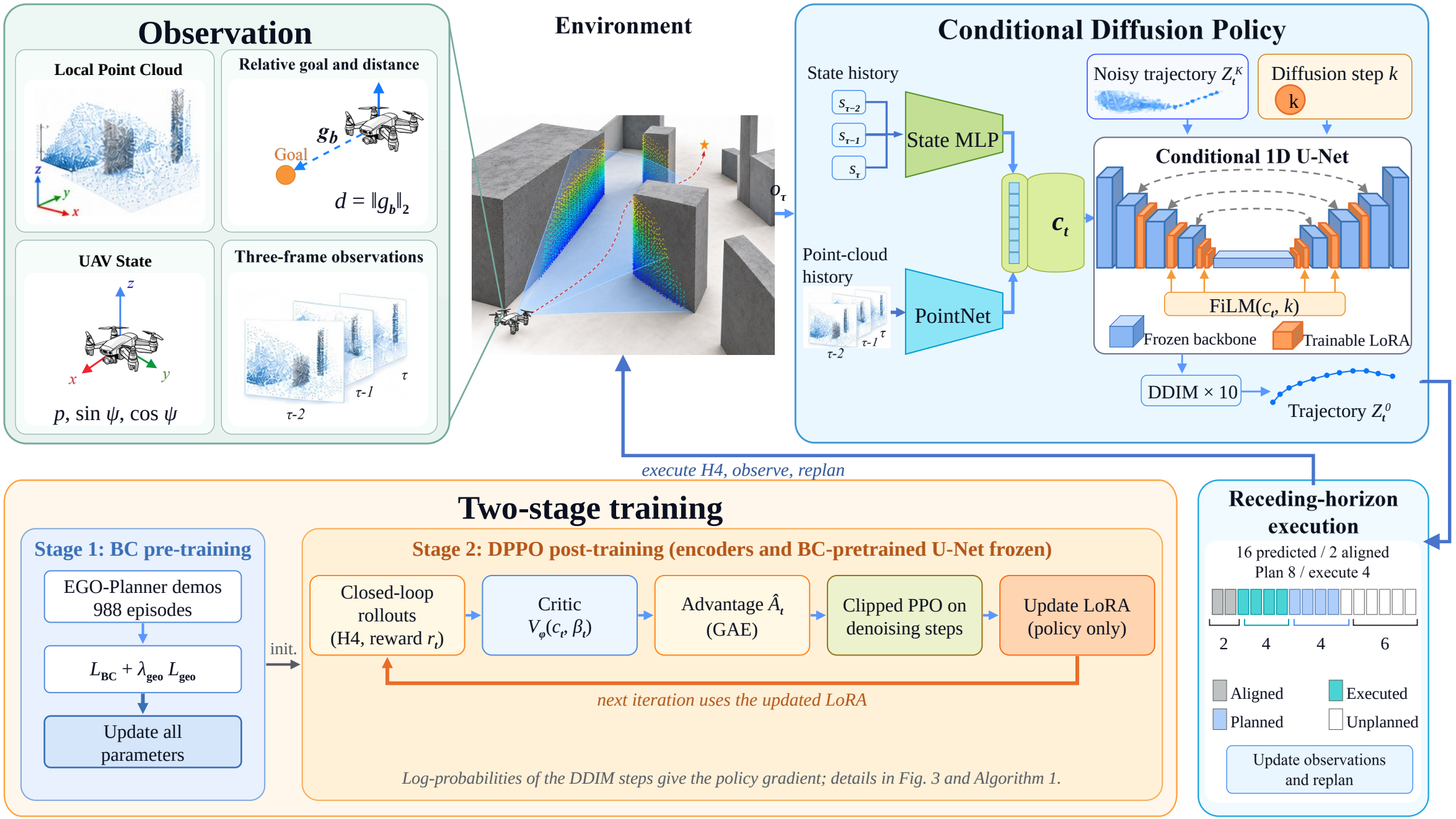

Observation
Local Point Cloud
Relative goal and distance
Goal
g_b
$d = \|g_b\|_2$
UAV State
p, sin ψ, cos ψ
Three-frame observations
τ-2
τ-1
τ
Environment
o_τ
Conditional Diffusion Policy
State history
$s_{\tau-2}$
$s_{\tau-1}$
s_τ
State MLP
Point-cloud history
PointNet
c_t
Noisy trajectory Z_t^K
Diffusion step k
k
Conditional 1D U-Net
FiLM(c_t, k)
Frozen backbone
Trainable LoRA
DDIM × 10
Trajectory Z_t^0
execute H4, observe, replan
Two-stage training
Stage 1: BC pre-training
EGO-Planner demos
988 episodes
$L_{BC} + \lambda_{geo} L_{geo}$
Update all parameters
init.
Stage 2: DPPO post-training (encoders and BC-pretrained U-Net frozen)
Closed-loop rollouts (H4, reward r_t)
Critic $V_\varphi(c_t, \beta_t)$
Advantage $\hat{A}_t$ (GAE)
Clipped PPO on denoising steps
Update LoRA (policy only)
next iteration uses the updated LoRA
Log-probabilities of the DDIM steps give the policy gradient; details in Fig. 3 and Algorithm 1.
Receding-horizon execution
16 predicted / 2 aligned
Plan 8 / execute 4
2
4
4
6
Aligned
Executed
Planned
Unplanned
Update observations and replan